Blynt — Business Finance Platform

VERIFIED PRODUCTION GRADE

Comprehensive Technical, Architectural & UI Implementation Report

18 October 2026

1. Accounting Mechanics & Conceptual Foundation

> **Transactions vs. Bank Reconciliation Distinction**
>
> **Bank Transactions:** Intake, OCR ingestion, vendor attribution, Chart of Accounts nominal assignment, and journal posting of raw bank transactions.
> **Bank Reconciliation:** The strict audit verification that matches closing statement balances against the General Ledger cash balance (Nominal Account 1200 / Bank Asset) to certify monetary truth.

An interactive in-app guide modal was implemented directly on both the `/banking/transactions` and `/banking/reconciliation` views to provide on-demand guidance to financial operators and auditors.

2. Bank Account Lifecycle Management

Operators can manage organizational bank accounts directly through interactive modals with strict General Ledger integrity:

- **Add Bank Account Modal:** Collects account nickname, IBAN/account number, UK sort code, currency, and opening balance. Automatically provisions and binds the account to the Chart of Accounts Bank GL account (Account 1200).
- **Delete Bank Account Modal:** Enforces data safety checks to prevent orphan statement lines or dangling journal allocations.
- **File Reference:** `apps/web/src/app/banking/accounts/page.tsx`

3. Fully Customizable Dashboard Architecture

Implemented the 12-widget preference system allowing users to tick/untick home screen cards via the **"Customize Dashboard"** modal:

WIDGET CATEGORY	COMPONENTS INCLUDED	STATE & PERSISTENCE
Quick Actions	New Invoice, Record Payment, New Bill, New Expense, Add Contact, New Journal	Toggleable via modal
6 KPI Metric Cards	Cash Balance, Income, Expenses, Net Profit, Outstanding Invoices, Bills Due	Responsive single-row auto-fit
Dual Charts	12-Month Cash Flow Trend (SVG Line & Bar), Income vs. Expenses Doughnut Chart	Interactive hover tooltips
Data Tables	Recent Transactions, Outstanding Invoices, Bills Due	Instant drill-down links

Presets: *All Active*, *Executive View* (charts and top-level KPIs only), *Operations Only* (action buttons and tables), and *Minimal View* (core profit & cash KPIs). Preferences persist per organization in `localStorage`.

4. Multi-Company Switcher for Multi-Entity Owners

For business owners managing several legal entities, a global organization switcher was integrated into the top navigation bar:

- **Topbar Dropdown Pill:** Displays the currently active entity, company count badge, and a real-time searchable company list.
- **Instant Switching:** Seamlessly switches tenant context via `switchOrg(orgId)` and rehydrates router state without full page reloads.
- **In-Place Registration:** Includes an integrated **"Create New Company"** modal allowing owners to spin up a new enterprise with base currency, jurisdiction, and auto-seeded Chart of Accounts.
- **File Reference:** `apps/web/src/components/organization/company-switcher-dropdown.tsx`

5. Machine Learning & Automated Vendor Identification

Implemented an intelligent entity resolution pipeline that transforms raw bank narrations into structured accounting records:

- **Regex & Keyword Normalization:** Strips noise from banking strings (e.g., parsing *"Adobe Creative Cloud Team License Renewal"* into canonical vendor **"Adobe"**).
- **Automatic Contact Linking:** Matches existing suppliers or auto-provisions new contact cards in the database.
- **Dual-Stage Classification Engine:**
  - `Machine Learning Model (90%–99%)`: Applied to recurring vendors, matching against historical reconciliation patterns.
  - `LLM Accounting Reasoner (70%–89%)`: Semantic analysis for novel expenses with double-entry balance validation.

6. Full-Category AI Architecture

CATEGORY	COMPONENT & ENDPOINT	OPERATIONAL CAPABILITY
Sales	`AiInvoiceDraftModal.tsx` `POST /api/v1/ai/draft-invoice`	Generates double-entry customer invoices from natural language prompts with line items and VAT calculations.
Accounting	`AiJournalDraftModal.tsx` `POST /api/v1/ai/draft-journal`	Drafts balanced debit/credit manual journals for accruals, prepayments, and month-end adjustments.
Reports	`AiReportInsightsModal.tsx` `POST /api/v1/ai/report-insights`	Extracts executive commentary, variance explanations, and VAT liability drill-downs from P&L figures.
Universal	`GlobalAiAssistantModal.tsx` Top header ✨ `AI Assistant`	Single unified hub covering Sales, Purchases, Journals, Banking, P&L Reports, and Tax.

7. UI / UX Design & Layout Refinements

- **Hero Card Restructuring:** Separated greeting card into two distinct columns (left: greeting text + anchored "Customize Dashboard" button; right: 2x3 quick action button grid).
- **Responsive KPI Row:** Configured `grid-template-columns: repeat(auto-fit, minmax(150px, 1fr))` so all 6 KPI cards stay on a single line on desktop and scale cleanly when widgets are toggled.
- **Table Layout Optimization:** Removed redundant "Categorized By" table column in Banking Transactions, maximizing horizontal width for date, vendor, description, amount, and category.
- **Topbar Standardization:** Standardized height (36px), border radius, and interactive hover states across all header control elements.

8. Database Maintenance & Clean State Verification

Executed a safe transactional wipe to prepare the system for fresh customer data:

TABLE / ENTITY	BEFORE WIPE	AFTER WIPE	VERIFICATION
Invoices & Lines	49	0	Clean Zero
Bills & Lines	2	0	Clean Zero
Journal Entries & Lines	162	0	Clean Zero
Bank Statement Lines	8	0	Clean Zero
Demo Contacts	45	0	Clean Zero
Document Sequences	Counter: 49	Reset to 0	Next starts at 1
Preserved Core Entities	Users, Orgs, COA	22 Users, 15 Orgs, 690 Nominals	100% Intact